

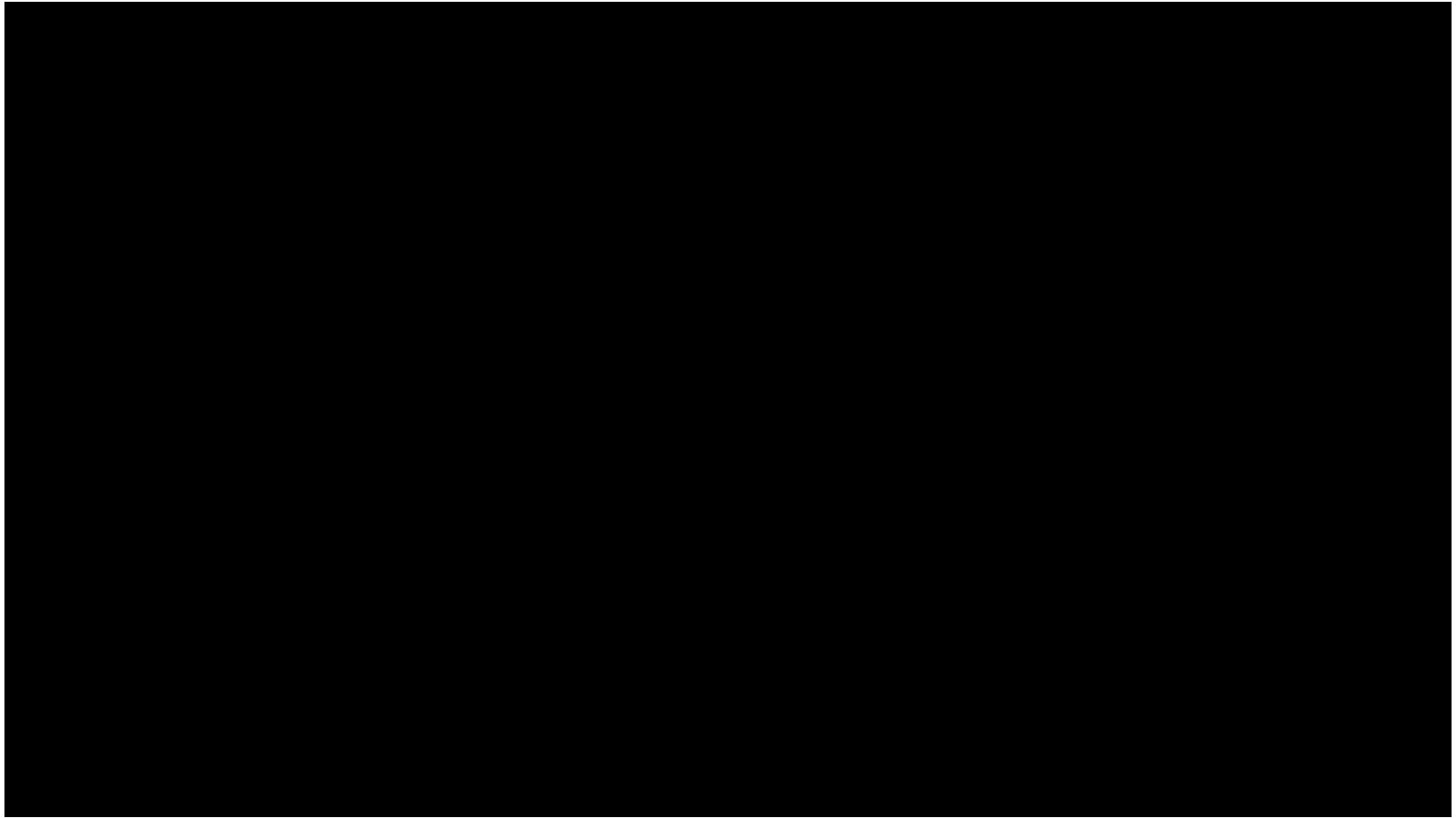


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Face recognition

What is face
recognition?

Face recognition



[Courtesy of Baidu]

Andrew Ng

Face verification vs. face recognition

→ Verification

- Input image, name/ID
- Output whether the input image is that of the claimed person

1:1

99.0%

99.9

→ Recognition

- Has a database of K persons
- Get an input image
- Output ID if the image is any of the K persons (or “not recognized”)

1:K

K=100 ←

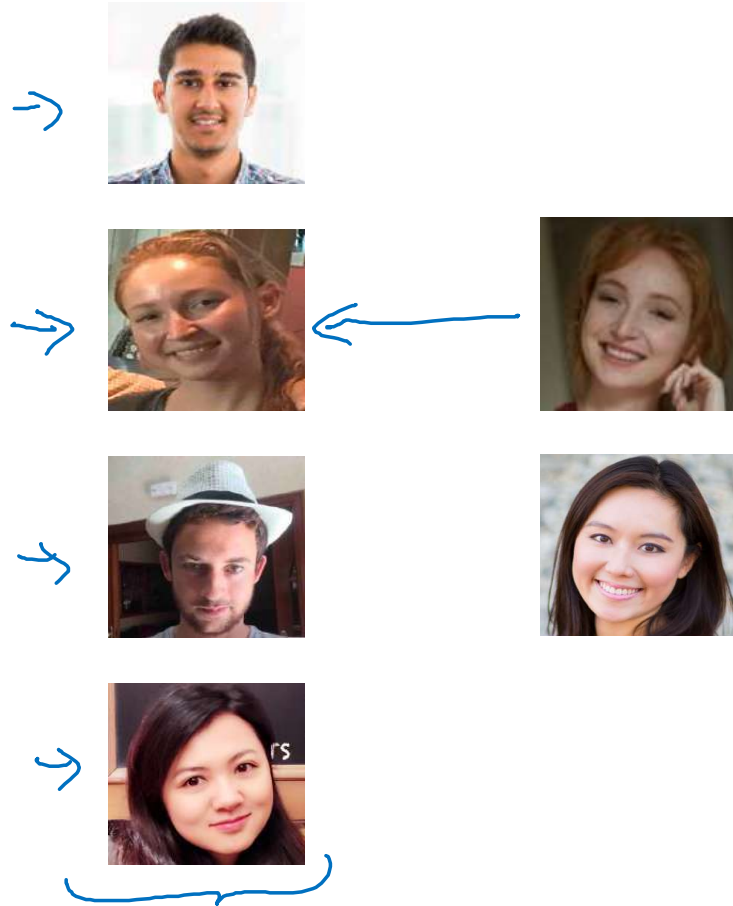


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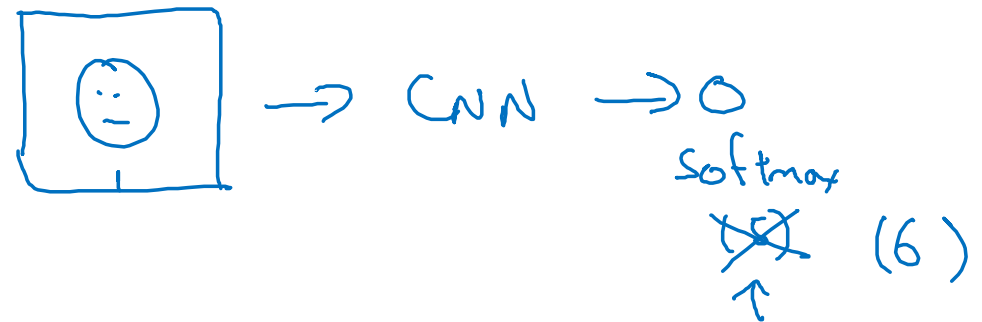
Face recognition

One-shot learning

One-shot learning



Learning from one example to recognize the person again



Learning a “similarity” function

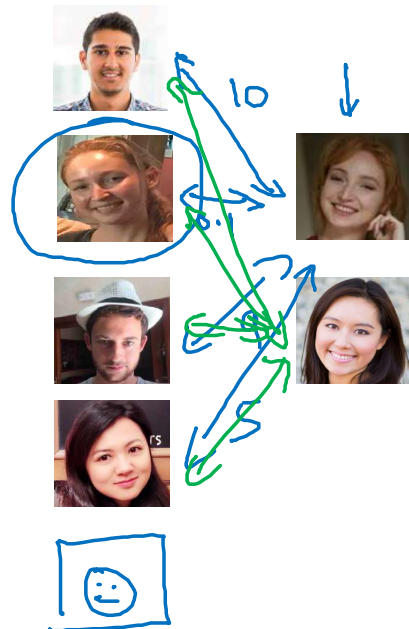
→ $d(\text{img1}, \text{img2})$ = degree of difference between images

If $d(\text{img1}, \text{img2}) \leq \tau$
 $> \tau$

“same”

“different”

} Verification.



$d(\text{img1}, \text{img2})$

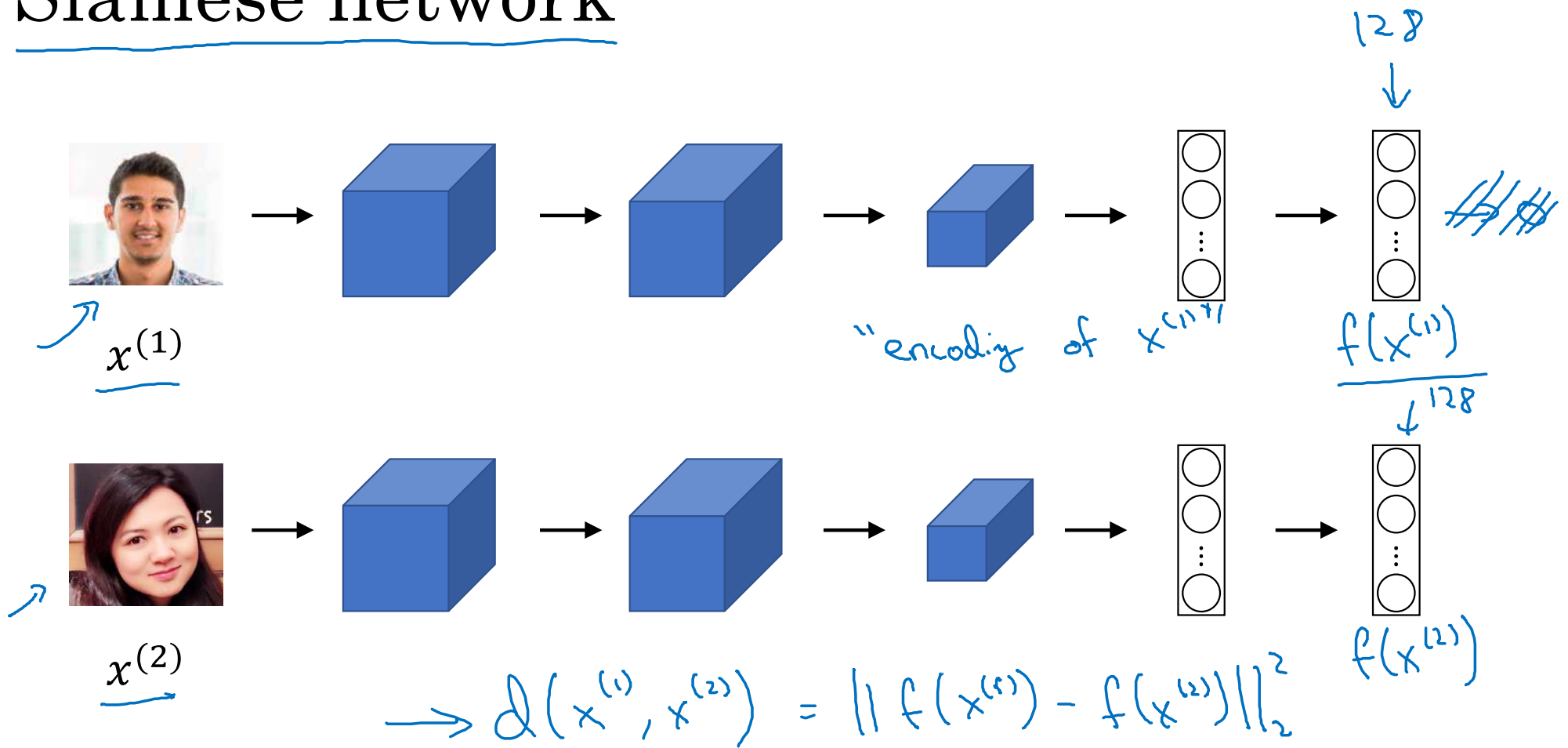


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Face recognition

Siamese network

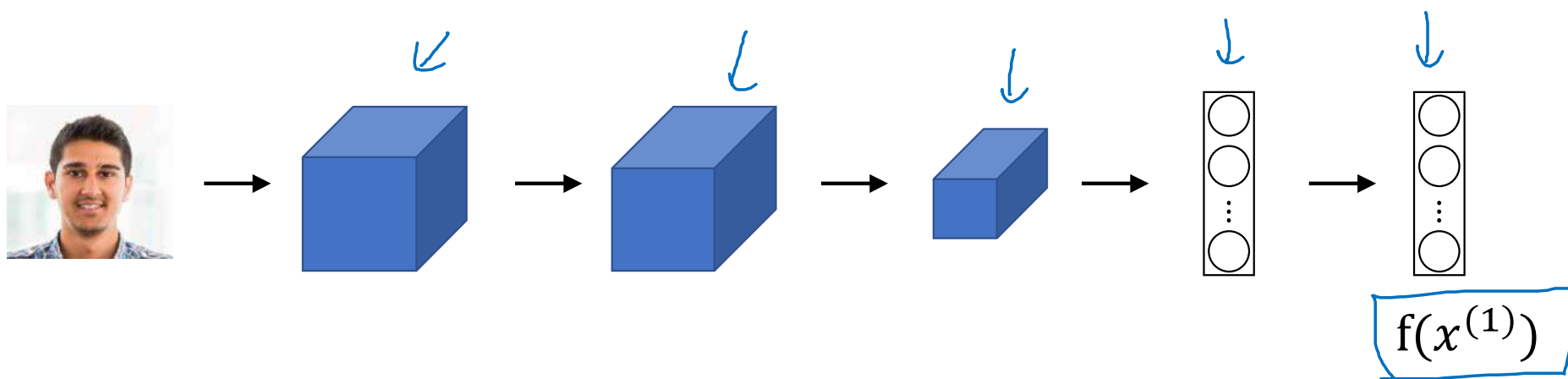
Siamese network



[Taigman et. al., 2014, DeepFace closing the gap to human level performance]

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Goal of learning



Parameters of NN define an encoding $f(x^{(i)})$

128

Learn parameters so that:

If $x^{(i)}, x^{(j)}$ are the same person, $\|f(x^{(i)}) - f(x^{(j)})\|^2$ is small.

If $x^{(i)}, x^{(j)}$ are different persons, $\|f(x^{(i)}) - f(x^{(j)})\|^2$ is large.

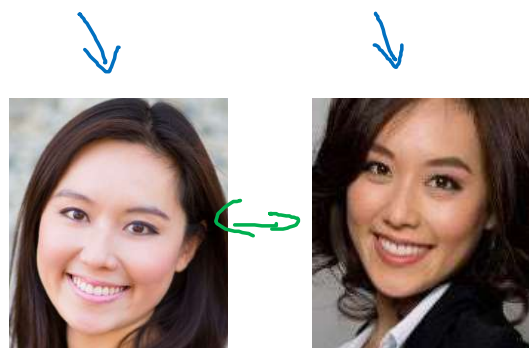


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Face recognition

Triplet loss

Learning Objective

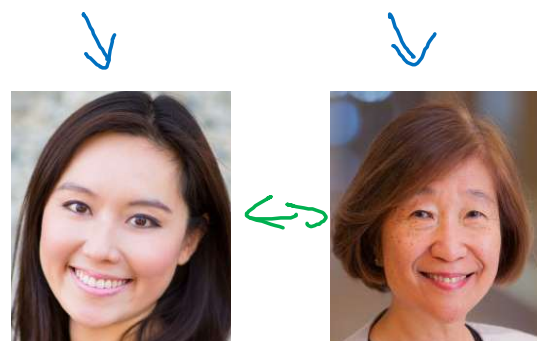


Anchor

Positive

A $d(A, P) = 0.5$

Want: $\underbrace{\|f(A) - f(P)\|^2}_{d(A, P)} + \underline{\alpha} \leq \overset{\nearrow 0.2}{}$



Anchor

Negative

A $d(A, N) = \cancel{0.5} = 0.7$

$\underbrace{\|f(A) - f(N)\|^2}_{d(A, N)}$

$$\underbrace{\|f(A) - f(P)\|^2}_0 - \underbrace{\|f(A) - f(N)\|^2}_0 + \underline{\alpha} \leq \underline{0} \quad \text{margin}$$

$f(\text{img}) = \vec{0}$

Loss function

Given 3 images A, P, N :

$$\underline{L(A, P, N)} = \max \left(\underbrace{\|f(A) - f(P)\|^2 - \|f(A) - f(N)\|^2 + \alpha}_{\geq 0}, 0 \right)$$

$$J = \sum_{i=1}^m L(A^{(i)}, P^{(i)}, N^{(i)})$$

A, P
↑ ↑

Training set: 10k pictures of 1k persons

Choosing the triplets A,P,N

During training, if A,P,N are chosen randomly,
 $d(A, P) + \alpha \leq d(A, N)$ is easily satisfied.

$$\|f(A) - f(P)\|^2 + \alpha \leq \|f(A) - f(N)\|^2$$

Choose triplets that're "hard" to train on.

$$\frac{d(A, P)}{d(A, P)} + \alpha \leq \frac{d(A, N)}{d(A, N)}$$

↓ ↑

Face Net
Deep Face

↓
[Schroff et al., 2015, FaceNet: A unified embedding for face recognition and clustering]

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Training set using triplet loss

Anchor



⋮



Positive



⋮



Negative



⋮



$$J$$
$$d(x^{(i)}, x^{(j)})$$

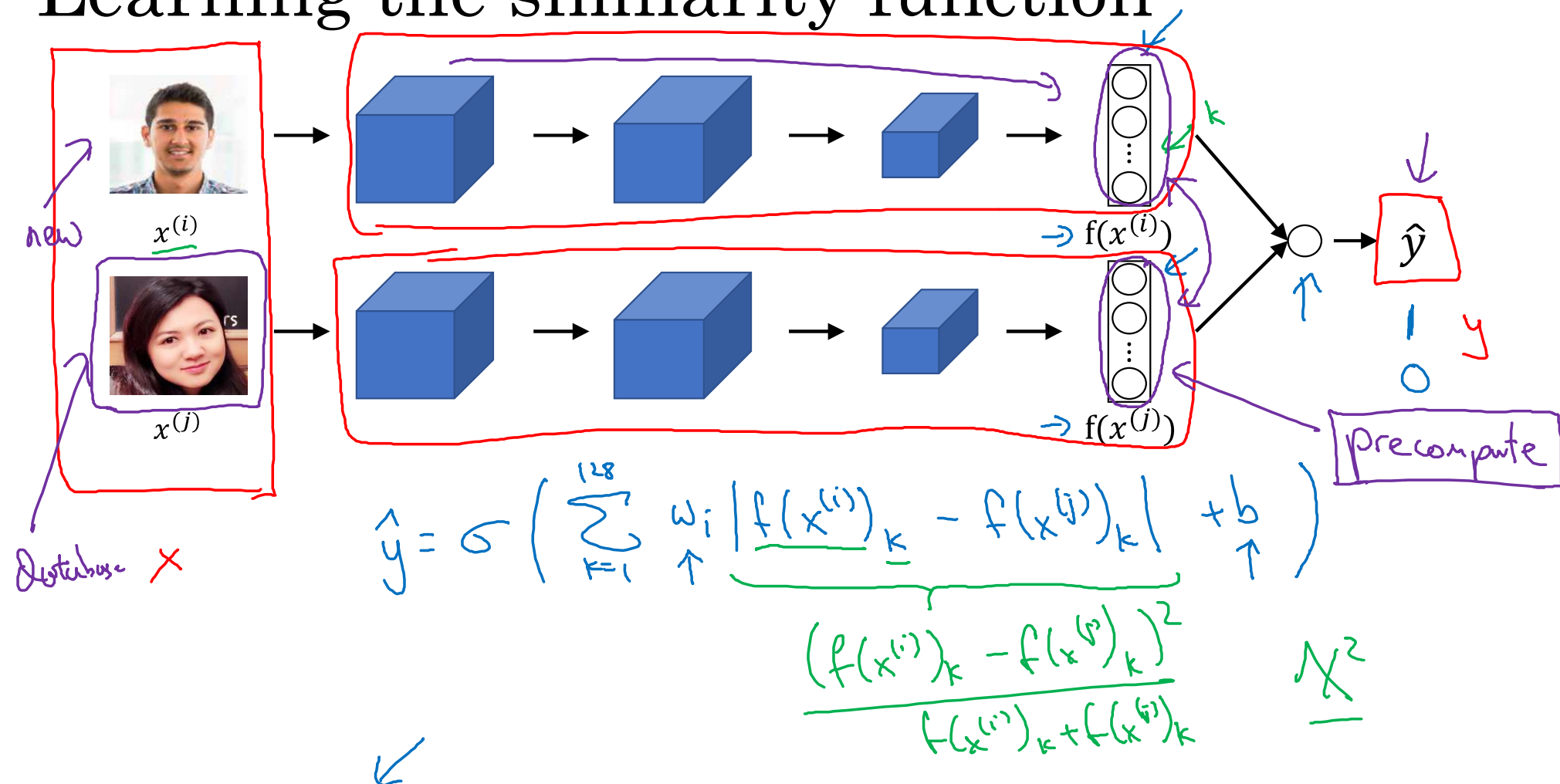


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Face recognition

Face verification and binary classification

Learning the similarity function



[Taigman et. al., 2014. DeepFace closing the gap to human level performance]

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Face verification supervised learning

x		y	
		1	"Same"
		0	"Different"
		0	
		1	

[Taigman et. al., 2014. DeepFace closing the gap to human level performance]

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Neural Style Transfer

What is neural style transfer?

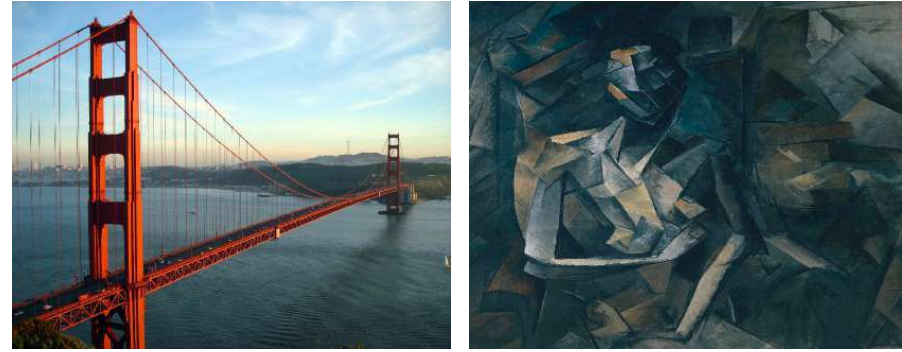
Neural style transfer



Content (C) Style (S)



Generated image (G)



Content (C) Style (S)



Generated image (G)

[Images generated by Justin Johnson]

Andrew Ng

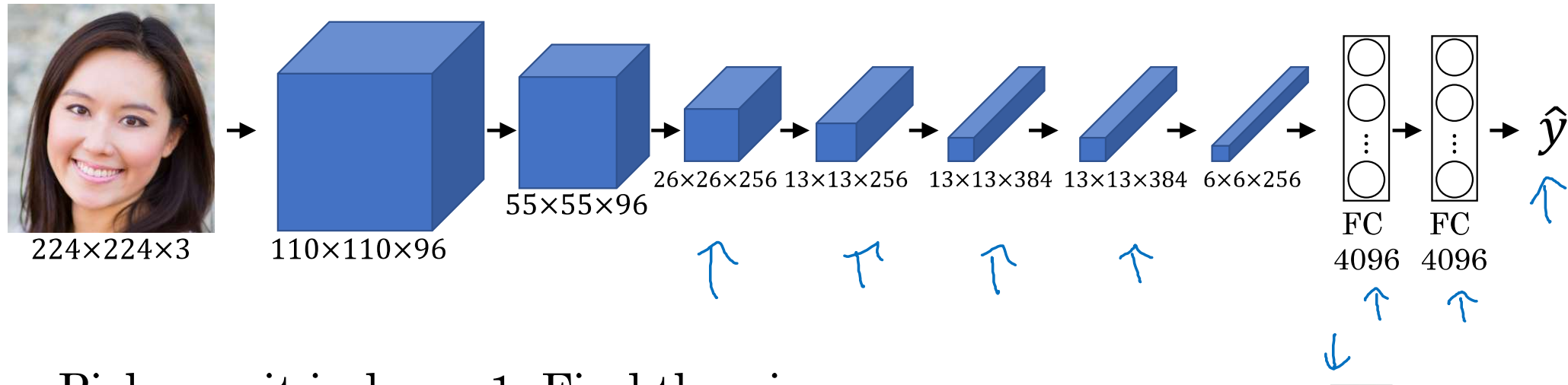


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Neural Style Transfer

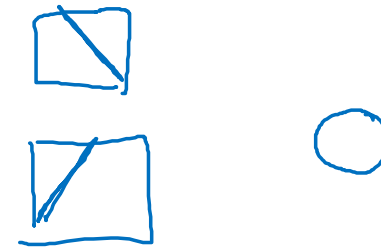
What are deep
ConvNets learning?

Visualizing what a deep network is learning

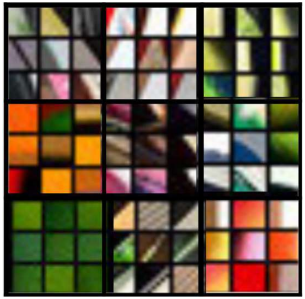


Pick a unit in layer 1. Find the nine image patches that maximize the unit's activation.

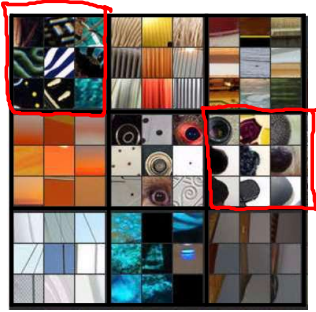
Repeat for other units.



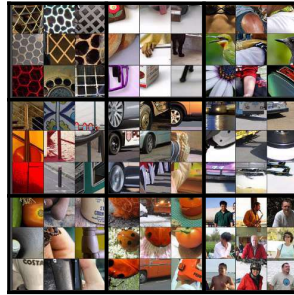
Visualizing deep layers



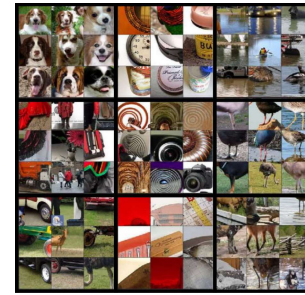
Layer 1



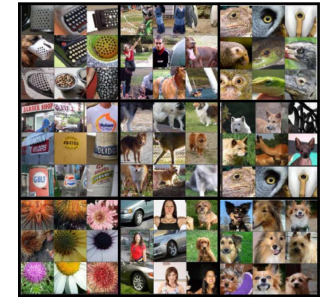
Layer 2



Layer 3

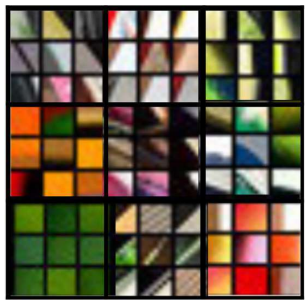


Layer 4



Layer 5

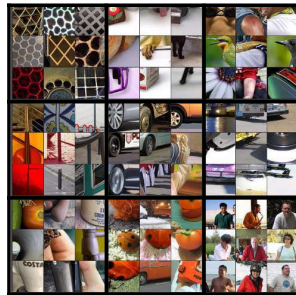
Visualizing deep layers: Layer 1



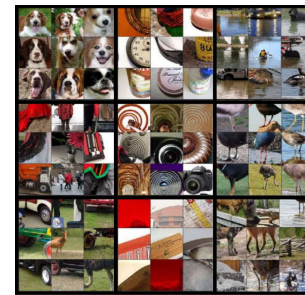
Layer 1



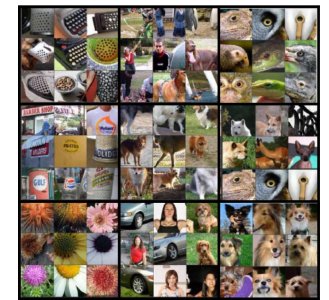
Layer 2



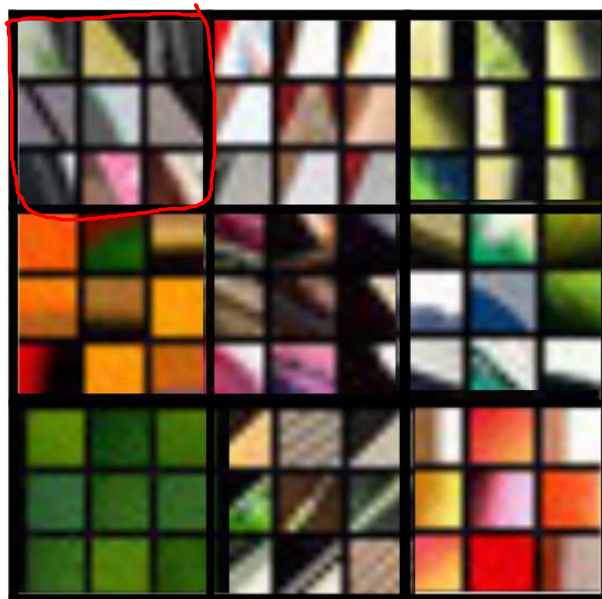
Layer 3



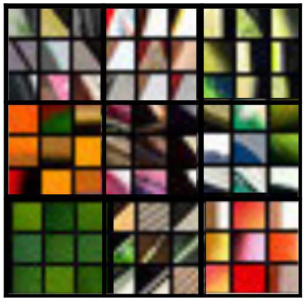
Layer 4



Layer 5



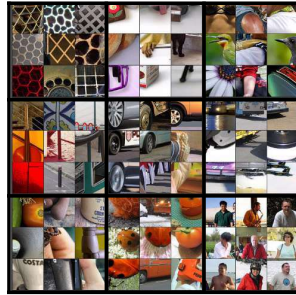
Visualizing deep layers: Layer 2



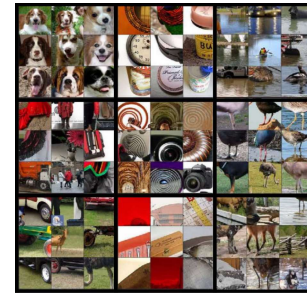
Layer 1



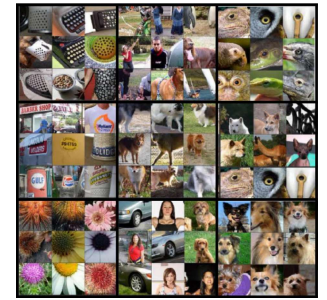
Layer 2



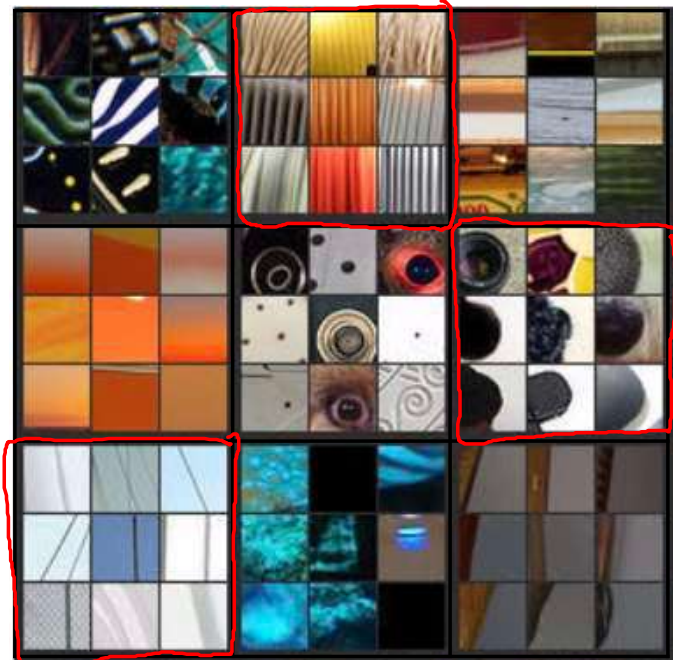
Layer 3



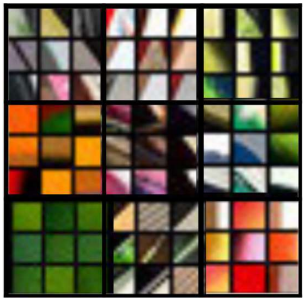
Layer 4



Layer 5



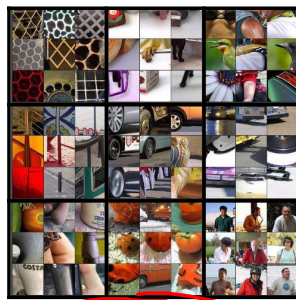
Visualizing deep layers: Layer 3



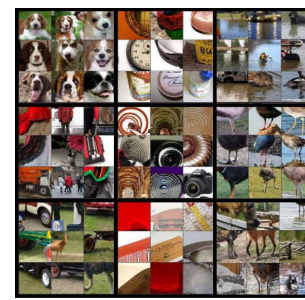
Layer 1



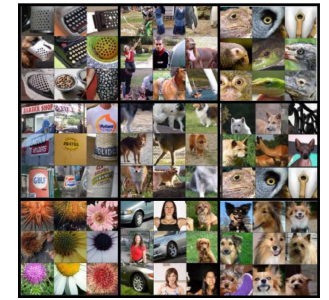
Layer 2



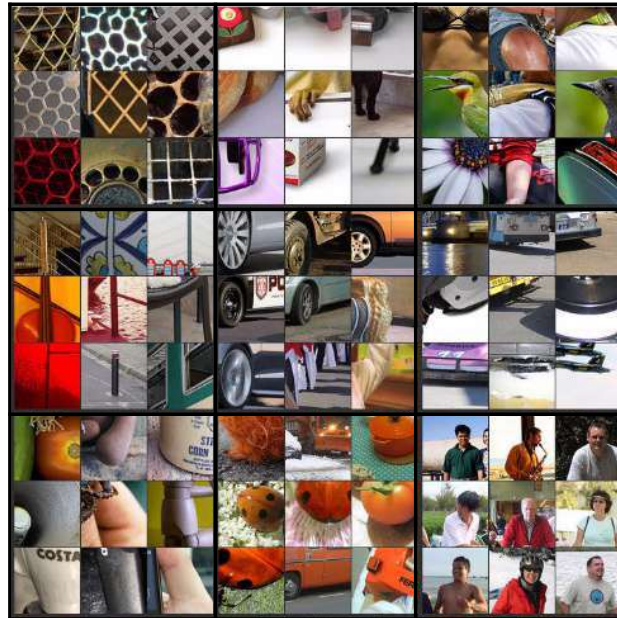
Layer 3



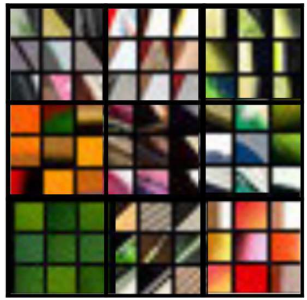
Layer 4



Layer 5



Visualizing deep layers: Layer 3

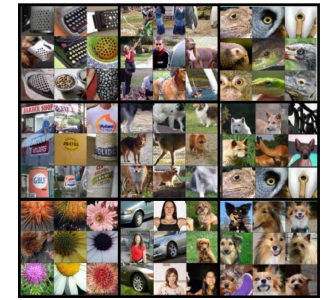


Layer 1



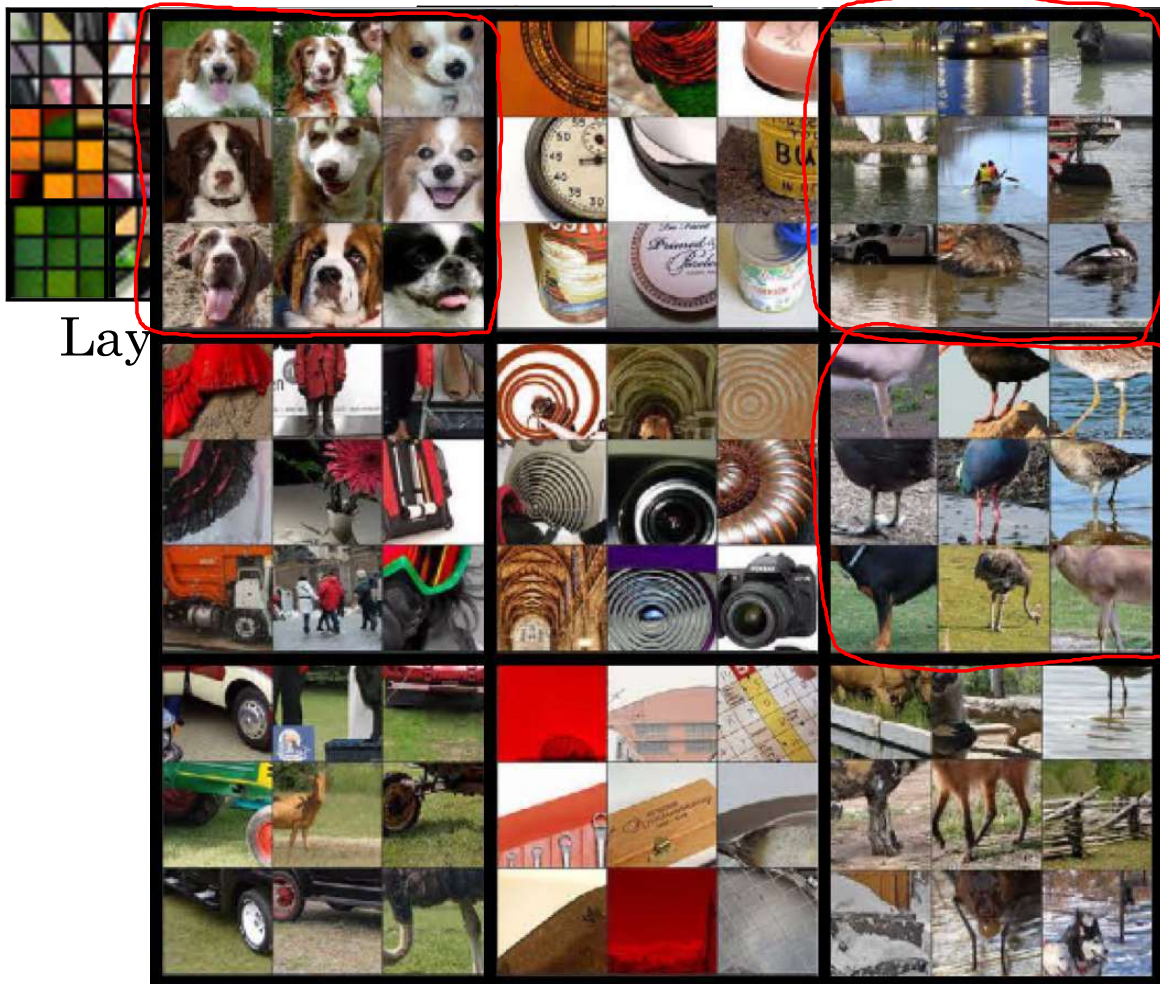
Layer 3

Layer 4

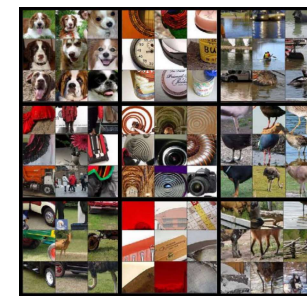


Layer 5

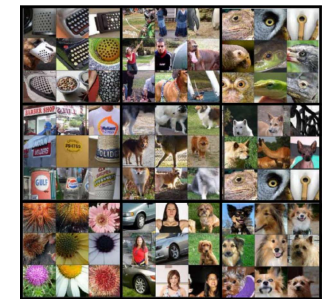
Visualizing deep layers: Layer 4



Layer 4

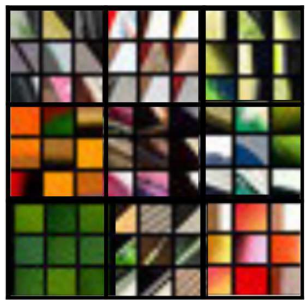


Layer 4

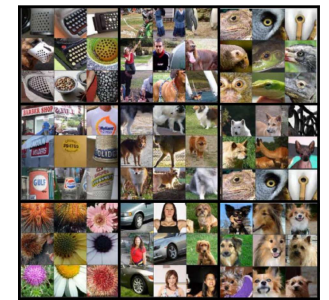


Layer 5

Visualizing deep layers: Layer 5



Layer 1



Layer 5



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Neural Style Transfer

Cost function

Neural style transfer cost function



Content C

Style S



Generated image G

$$\mathcal{J}(G) = \alpha \mathcal{J}_{\text{content}}(\overset{\text{C}}{\underbrace{\quad}}, \overset{\text{G}}{\underbrace{\quad}}) + \beta \mathcal{J}_{\text{style}}(\underset{\text{S}}{\underbrace{\quad}}, \underset{\text{G}}{\underbrace{\quad}})$$

Find the generated image G

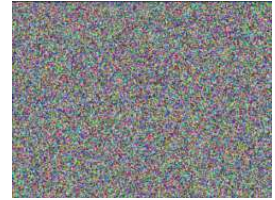
1. Initiate G randomly

G : $100 \times 100 \times 3$

↑
RGB

2. Use gradient descent to minimize $J(G)$

$$G := G - \frac{\partial}{\partial G} J(G)$$





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Neural Style Transfer

Content cost function

Content cost function

$$\underline{J(G)} = \alpha \underline{J_{content}(C, G)} + \beta J_{style}(S, G)$$

- Say you use hidden layer l to compute content cost.
- Use pre-trained ConvNet. (E.g., VGG network)
- Let $a^{[l](C)}$ and $a^{[l](G)}$ be the activation of layer l on the images
- If $a^{[l](C)}$ and $a^{[l](G)}$ are similar, both images have similar content

$$J_{content}(C, G) = \frac{1}{2} \left\| \underbrace{a^{[l](C)}}_{\text{activation of layer } l \text{ on image } C} - \underbrace{a^{[l](G)}}_{\text{activation of layer } l \text{ on image } G} \right\|^2$$

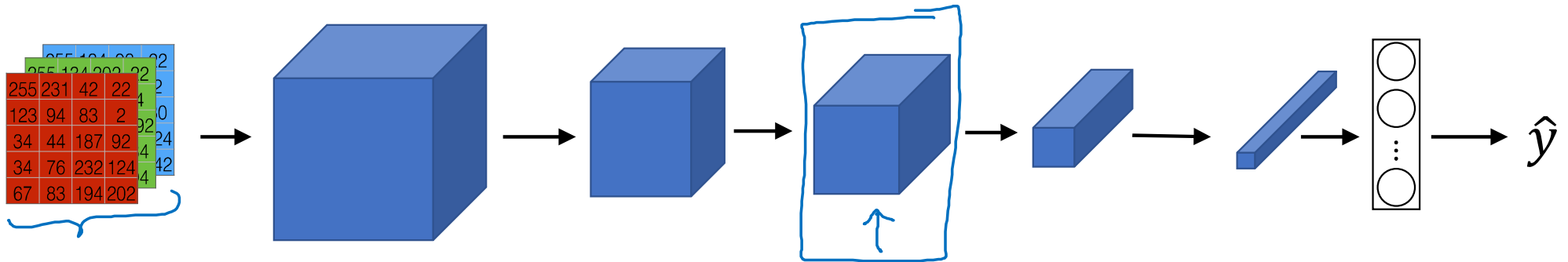


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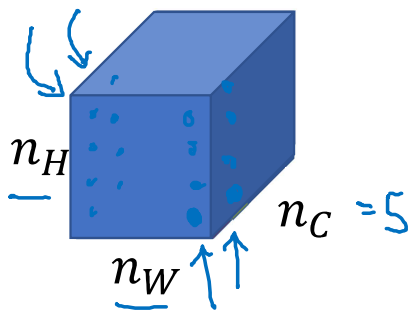
Neural Style Transfer

Style cost function

Meaning of the “style” of an image



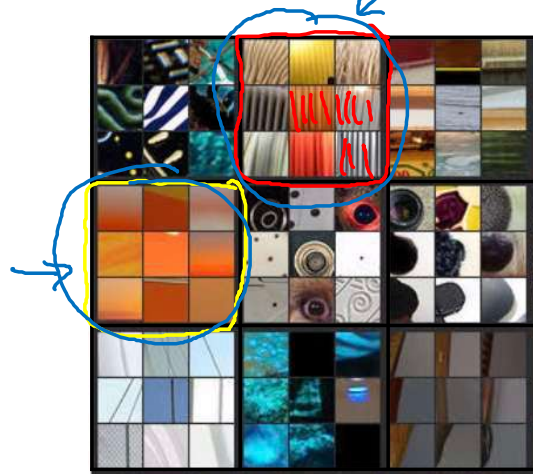
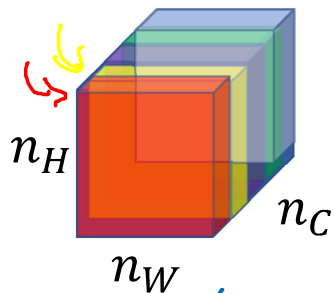
Say you are using layer l 's activation to measure “style.”
Define style as correlation between activations across channels.



How correlated are the activations
across different channels?

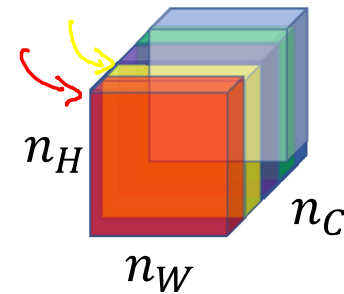
Intuition about style of an image

Style image



Correlated?
Uncorrelated

Generated Image



Style matrix

Let $\underline{a}_{i,j,k}^{[l]}$ = activation at (i, j, k) . $\underline{G}^{[l]}$ is $\underline{n}_c^{[l]} \times \underline{n}_c^{[l]}$

$$\begin{aligned} \rightarrow \underline{G}_{kk'}^{[l](s)} &= \sum_{i=1}^{n_H^{[l]}} \sum_{j=1}^{n_W^{[l]}} \underline{a}_{ijk}^{[l](s)} \underline{a}_{ijk'}^{[l](s)} \\ \rightarrow \underline{G}_{kk'}^{[l](G)} &= \sum_{i=1}^{n_H^{[l]}} \sum_{j=1}^{n_W^{[l]}} \underline{a}_{ijk}^{[l](G)} \underline{a}_{ijk}^{[l](G)} \end{aligned}$$

"Gram matrix"

$$\begin{aligned} & n_c \\ & \underline{G}_{kk'}^{[l]} \\ & \uparrow \downarrow \\ & k = 1, \dots, n_c^{[l]} \end{aligned}$$

$$\begin{aligned} \uparrow \beta \quad J_{\text{style}}^{[l]}(S, G) &= \frac{1}{(\dots)} \left\| \underline{G}^{[l](s)} - \underline{G}^{[l](G)} \right\|_F^2 \\ &= \frac{1}{(2n_H^{[l]}n_W^{[l]}n_c^{[l]})^2} \sum_k \sum_{k'} \left(\underline{G}_{kk'}^{[l](s)} - \underline{G}_{kk'}^{[l](G)} \right)^2 \end{aligned}$$

Style cost function

$$\|G^{TJ(S)} - G^{TJ(G)}\|_F^2$$

$$J_{style}^{[l]}(S, G) = \frac{1}{\left(2n_H^{[l]}n_W^{[l]}n_C^{[l]}\right)^2} \sum_k \sum_{k'} (G_{kk'}^{[l](S)} - G_{kk'}^{[l](G)})$$

$$J_{style}(S, G) = \sum_l \lambda_l^{TJ} J_{style}^{TJ}(S, G)$$

$$\underbrace{J(G)}_G = \alpha J_{content}(G) + \beta J_{style}(S, G)$$

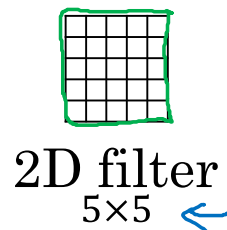
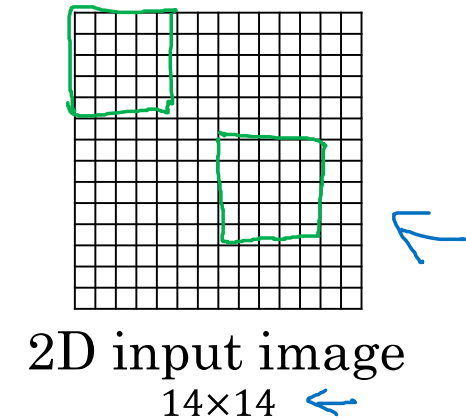
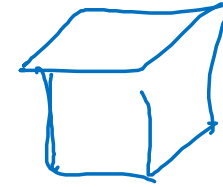


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Convolutional Networks in 1D or 3D

1D and 3D
generalizations of
models

Convolutions in 2D and 1D



*

$$14 \times 14 \times \underline{3} * 5 \times 5 \times \underline{3}$$

$$\rightarrow \underline{10 \times 10 \times 16}$$

$$10 \times 10 \times \underline{16} * 5 \times 5 \times \underline{16}$$

$$\rightarrow \underline{6 \times 6 \times 32}$$



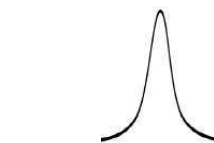
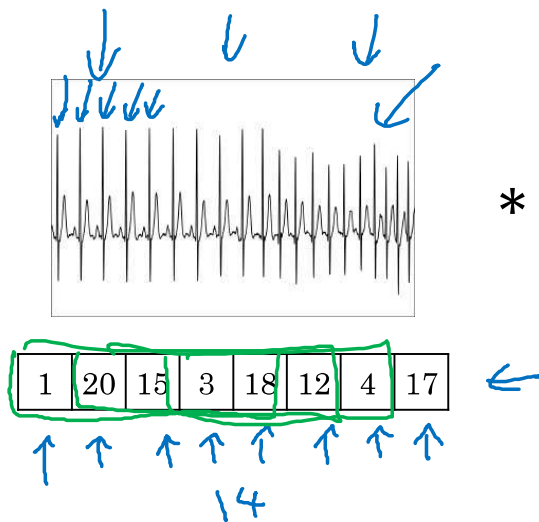
$$14 \times \underline{1} * 5 \times \underline{1}$$

$$\rightarrow \underline{10 \times 16}$$

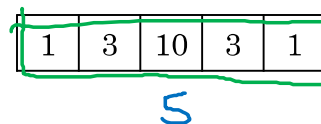
$$10 \times \underline{16} * 5 \times \underline{16}$$

$$\rightarrow \underline{6 \times 32}$$

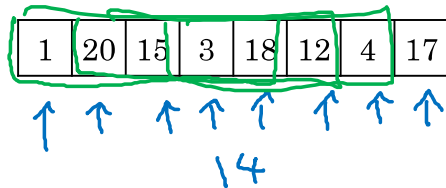
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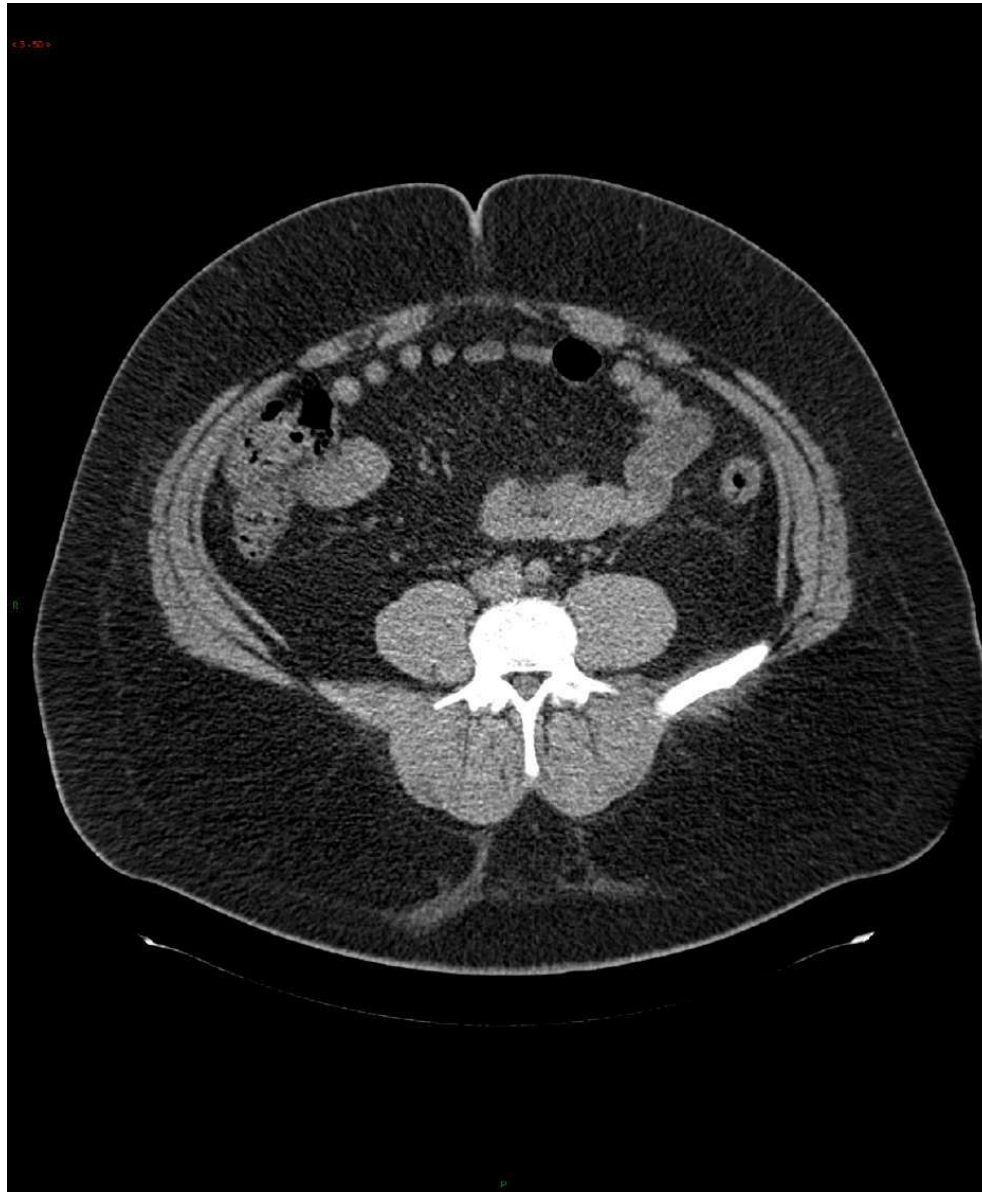
*



←



3D data



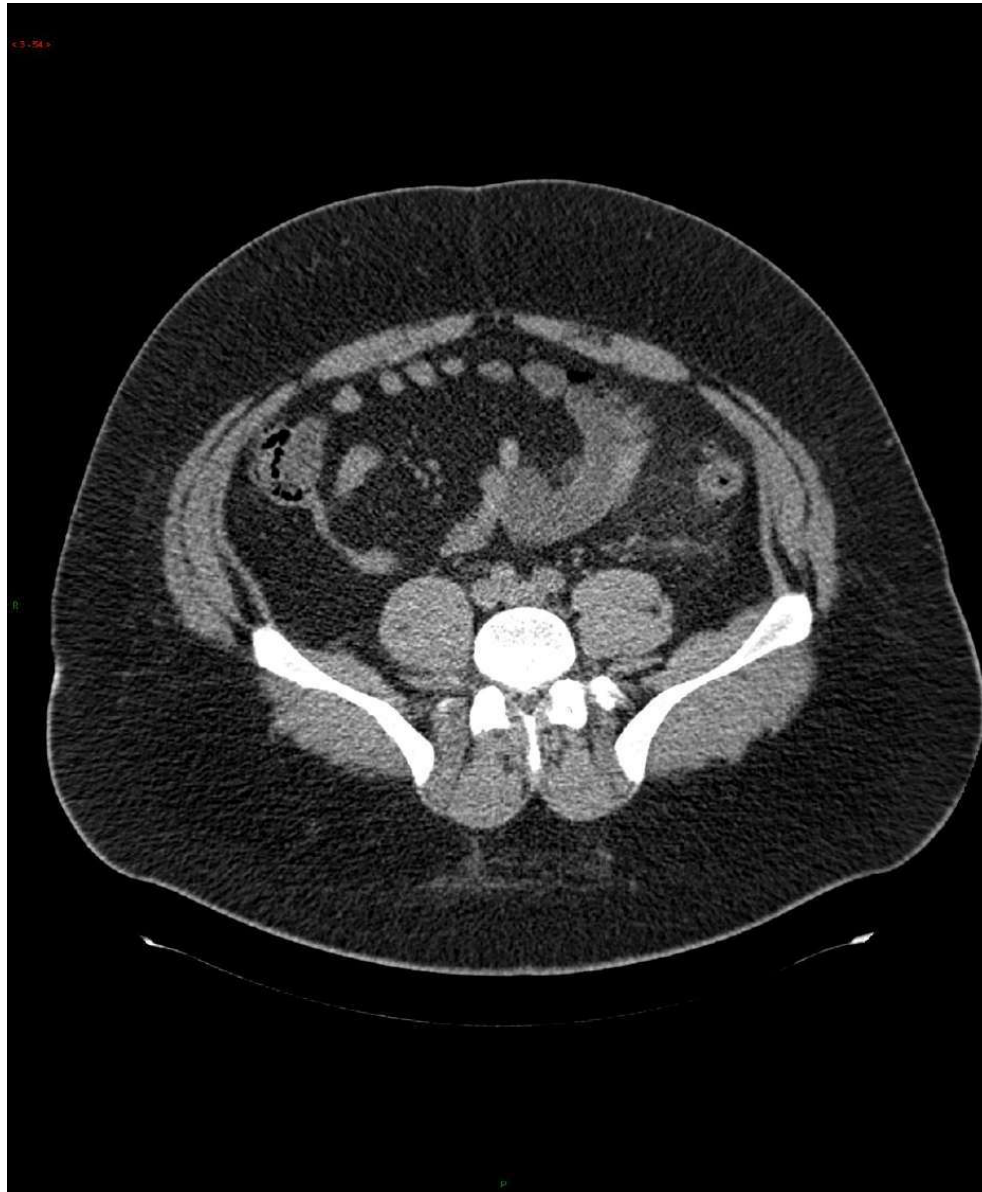
Andrew Ng

3D data



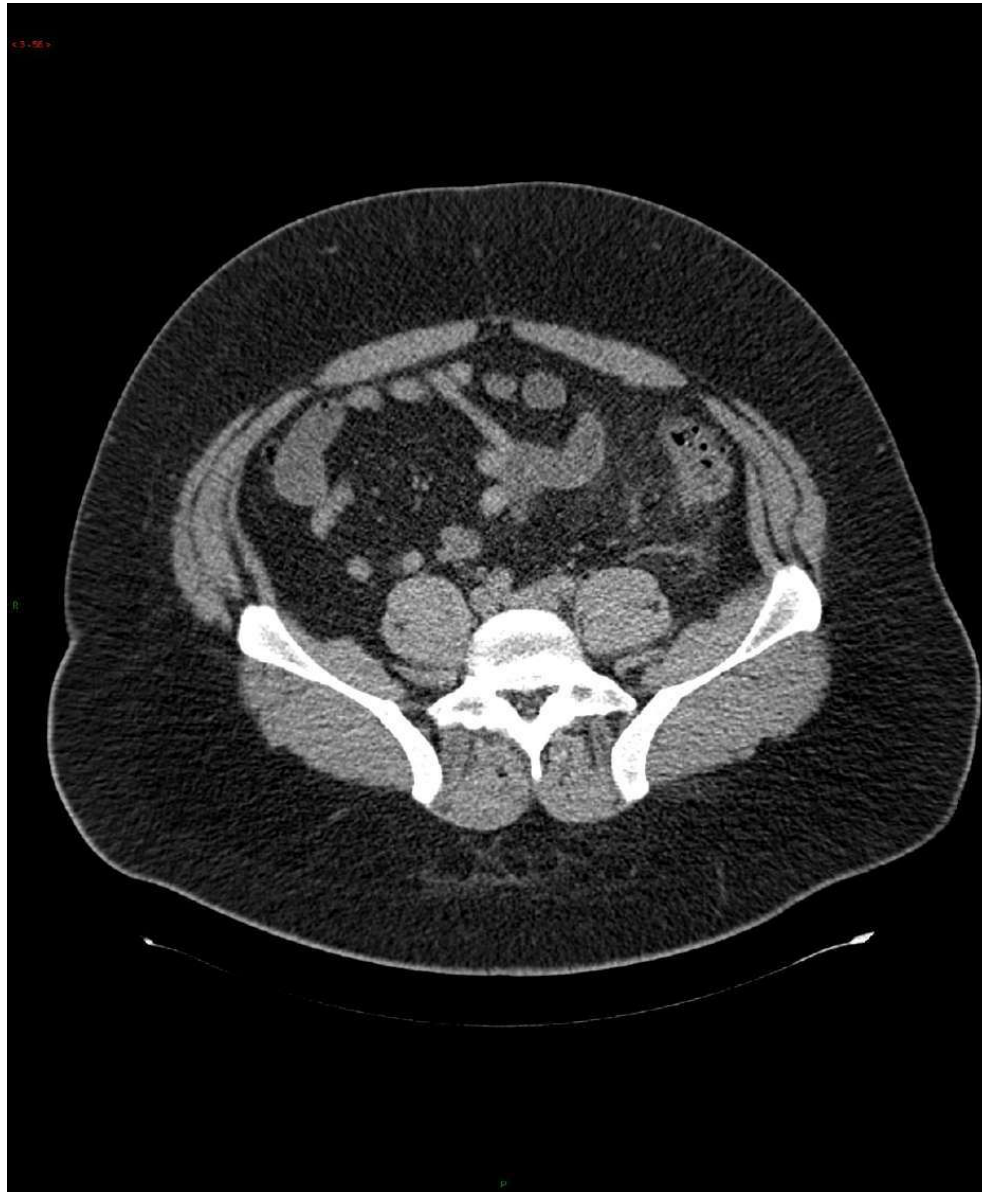
Andrew Ng

3D data



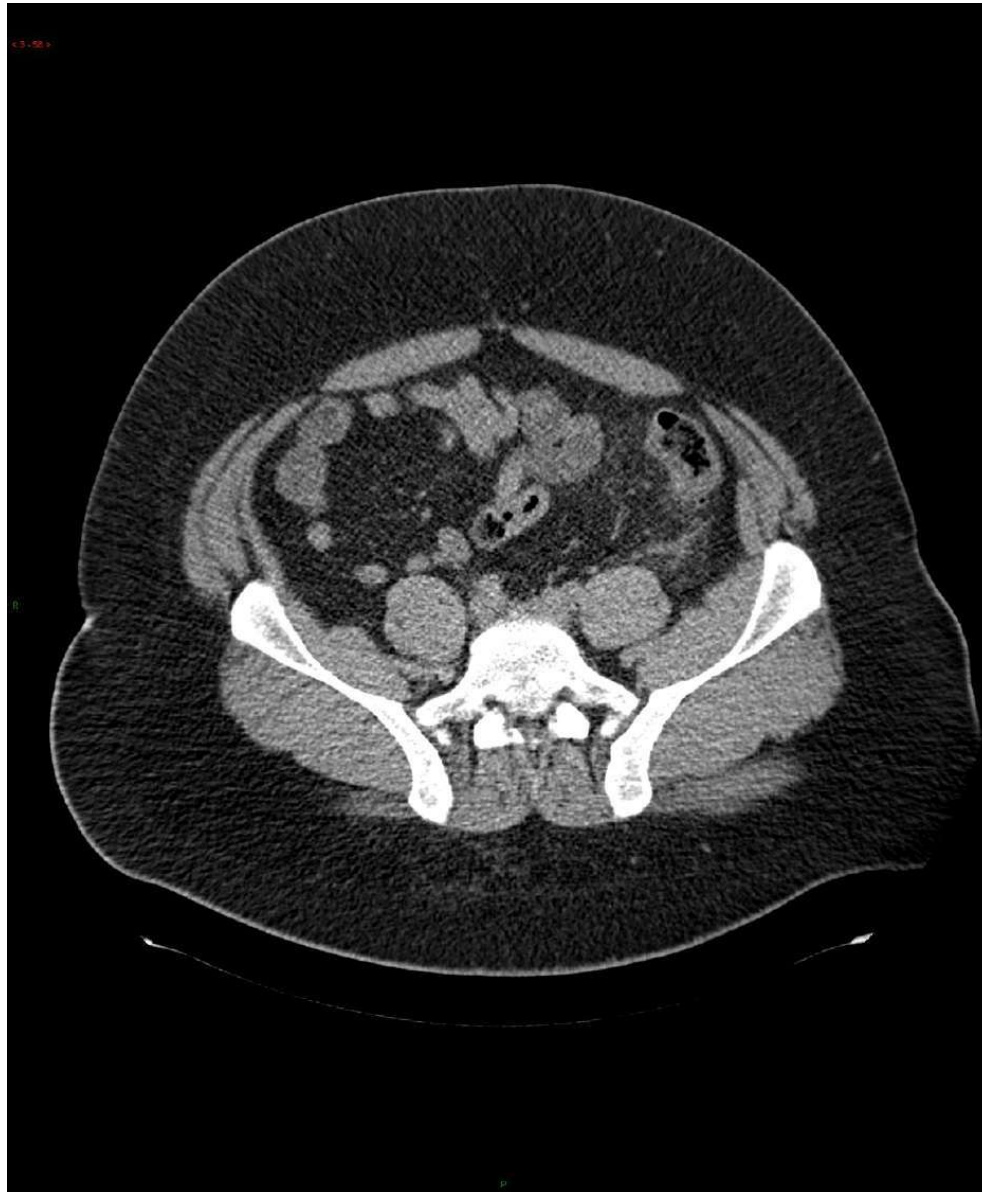
Andrew Ng

3D data



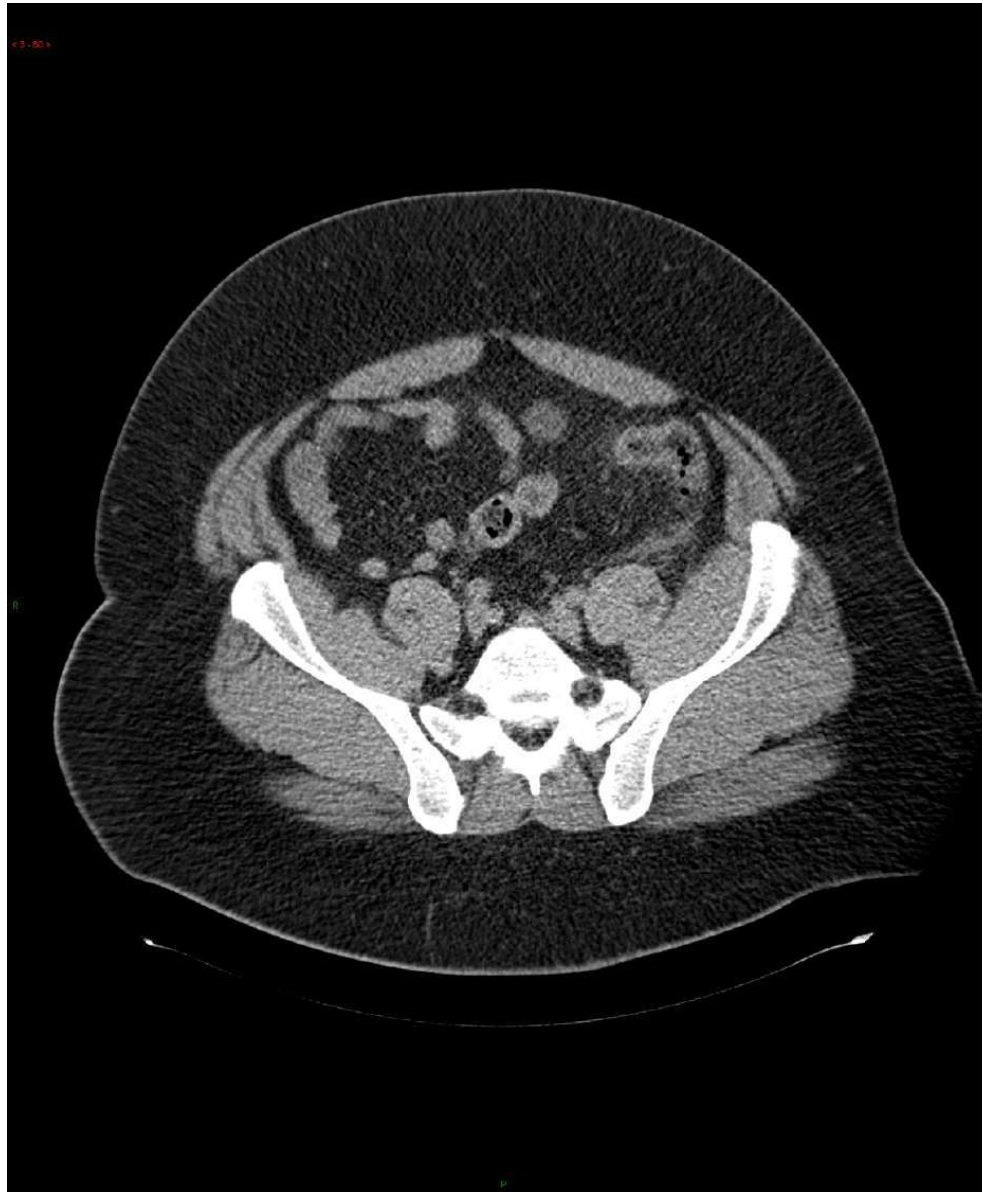
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3D data



Andrew Ng

3D data



Andrew Ng

3D data



Andrew Ng

3D data



Andrew Ng

3D data



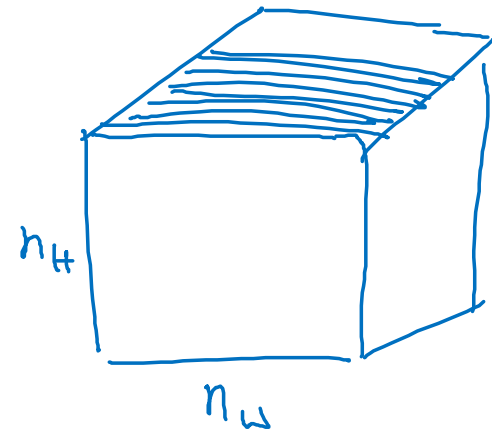
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3D data



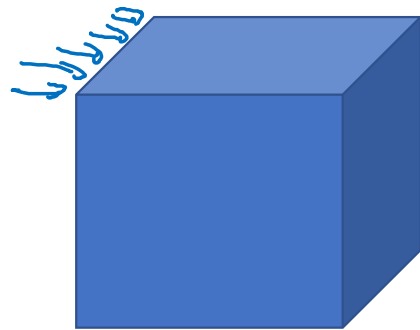
Andrew Ng

3D data



Andrew Ng

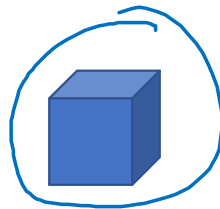
3D convolution



3D volume



*



3D filter

$$\begin{array}{l}
 \downarrow \quad \downarrow \quad \downarrow \quad \downarrow n_c \\
 \underline{14 \times 14 \times 14} \times \underline{1} \\
 * \quad \underline{5 \times 5 \times 5} \times \underline{1} \quad 16 \text{ filters} \\
 \rightarrow 10 \times 10 \times 10 \times \underline{16} \\
 * \quad \underline{5 \times 5 \times 5} \times \underline{16} \quad 32 \text{ filters} \\
 \rightarrow 6 \times 6 \times 6 \times 32
 \end{array}$$